COURSE DESCRIPTION FORM

INSTITUTION National University of Computer and Emerging Sciences (NUCES-FAST)

PROGRAM (S) TO BE EVALUATED BS (SE, CS,AI,CY)

A. Course Description

Course Code	CS4045
Course Title	Deep Learning for Perception
Credit Hours	3
Prerequisites by Course(s) and Topics	Data Structures
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Mid-I: 15 Mid-II: 15 Assignments: 10 Project: 10 Final: 50
Course Coordinator	Dr. Jawwad A Shamsi
URL (if any)	
Current Catalog Description	Introduction to Machine Learning and Deep Learning. Supervised Learning, Unsupervised Learning, Regression vs classification, Clustering Reinforcement Learning. Bayesian machine learning models, Ensemble methods, Random forest, Introduction to Neural Networks, Why Neural Networks, Shallow vs Deep Neural Networks, Neurons, Perceptron learning algorithm, weights, bias and variance, Training / Testing datasets, Activation functions sigmoid, RELU, tanh, layers, input layers, dense layers, softmax layers, max pooling, average pooling hidden layers, output layers, Loss functions, Practical aspects of deep learning, Vanishing Gradient , Batch normalization, stride, pooling, Optimization, and programming frameworks (Keras and Pytorch), Structuring Deep learning projects, practical examples and case studies, Convolutional neural networks, Computer vision examples, Resnets, Skip connections, Ensemble Learning, Inception net, Transfer learning, Object detection, anchor boxes, Region proposal Networks, IoU, YOLO, Sequence models, LSTM, Time series prediction, GRU, Attention mechanism, Machine Translation, Auto-Encoders, Variational Auto Encoders, Generative Adversial Networks (GANs), Deep learning: Hype and reality, small sample size, Diffusion Models, LLMS, Agentic AI, Other issues and future prospects
Textbook (or **Laboratory Manual**	Instructor's own lecture notes (Text)

for Laboratory Courses)	Reference: 1) Deep Learning for Vision Based Systems https://livebook.manning.com/book/deep-learning-for-vision-systems/welcome/v-8/ 2) Dive into deep learning https://d2l.ai/ 3) I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning, MIT Press, 2016
Reference Material	1) Yaser S. Abu Mustafa and Malik Magdoon Ismail, Learning from Data 2) Dive into Deep Learning.
Course Goals	

A. Course Learning Outcomes (CLOs)

1. To impart fundamental concepts about Deep Learning and Machine Learning
2. To study and apply various applications and case studies of deep learning in various domains.
3. To develop hands-on skills on well-established deep neural network architectures and deep learning software frameworks
4. To study directions for future research in deep learning

B. Program Learning Outcomes

For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.

1. Academic Education:	To prepare graduates as computing professionals	✔
2. Knowledge for Solving Computing Problems:	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.	✔
3. Problem Analysis:	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.	
4. Design/ Development of Solutions:	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✔
5. Modern Tool Usage:	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an	✔

	understanding of the limitations.	
6. Individual and Teamwork:	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.	✔
7. Communication:	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.	✔
8. Computing Professionalism and Society:	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.	✔
9. Ethics:	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.	✔
10. Life-long Learning:	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.	✔

C. Relation between CLOs and PLOs
(CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)

		PLOs / GAs									
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
CLOs	1	✔	✔								
	2	✔	✔	✔	✔	✔	✔				
	3	✔	✔	✔	✔	✔	✔	✔			
	4	✔	✔		✔		✔		✔	✔	✔

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)

1. Topics to be covered:

Week #	Topics to be Covered	Contact hours	CLO
1	Introduction of Course. Fundamentals, AIvs ML,vs DL	03	1

		ML Types Classification, Clustering, Regression, Confusion Matrix Machine Learning Basics Feature Engineering, Distance Calculation		
	2	Introduction to DNN, weights and bias, matrix computation Neuron, Hidden Layers, Activation Functions, DNN vs NN, Linearity vs Non Linearity, OR gate	03	1,2
	3	Forward Propagation, Backward Propagation **Logic Gates Assignment**, Back Propagation with numerical example	03	1,2,3
	4	Intro to CNN, pooling, filters, paramaeters, convolution, types of filters, RELU Intro to CNN, pooling, filters, paramaeters	03	1,2
	5	**CNN lab Assignment** Overfitting, no. features in CNN, Dropout, Vanising Gradient Problem, Normalization, Resnet Facenet Siamese Network, Triplet Loss	03	1,2,3
	6	Sessional - 1	01	
	7	Inception Network Object Classification. Intro to Yolo, **Yolo Assignment**	03	2,3
	8	Yolo. Sequence to Sequence Networks RNN LSTM Padding and Parameter Sharing in RNNs	03	1,2,3
	9	GRU and Bi-RNN. Intro to NLP, Word Embeddings, word2vec	03	1,2,4
	10	Word Embeddings contd. Auto Encoders, VAEs. **Assignment**	03	2,3

	11	Sessional II	01	
	12	Generative Adversarial Networks/ Deep Fakes	03	04
	13	Reinforcement Learning. **Assignment**	03	04
	14	Transformers and LLMs	03	03,04
	15	LLMs contd. Diffusion Models and Intro to Agentic AI	03	03,04
	16	Revision and course projects	03	03,04
		Total	46	

Laboratory Projects/Experiments Done in the Course	Hands on experience / Tutorials on implementation of deep learning models			
Programming Assignments Done in the Course	We will have programming exercises in alternate weeks			
Class Time Spent on (in credit hours)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**
	35	30	30	5
Oral and Written Communications	Every student / group is required to design a project using deep leaning models. The assessment of the project will be based on innovation, practicality, and clarity. Each group will demonstrate the project to the instructors and peers, deliver a 10 mins project presentation and a well written project report.			

Instructor Name: **Dr. Jawwad A Shamsi**

Instructor Signature: _____

Date January 21st 2026